

Partial results toward the two-base integer-exponent conjecture

A survey of the machine-checked corpus in **ProveIt**

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Abstract

We survey the unconditional partial results, all formally verified in Lean 4, surrounding the conjecture that a real number x with $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$ must itself be an integer. The conjecture is a special case of the four exponentials conjecture and is open; the three-base analogue (with $5^x \in \mathbb{Z}$ added) is a theorem, formalized via Waldschmidt’s interpolation-determinant proof. The corpus establishes: a verified zero-free region $2^x < 4096$ (so any nonintegral solution has $x \geq 12$); transcendence of every hypothetical nonintegral solution (via Gelfond–Schneider); a multiplicative rank-two theorem with a common odd core — every integral value 2^x has the form $2^i w^j$ for one fixed odd $w > 1$ — and the counting bound $\log_2 N \cdot \log_3 N$ for candidate values below N , linear in each dyadic block; a local three-term arithmetic-progression obstruction giving zero density via Roth numbers; exact fractional-part gaps and effective rational-approximation lower bounds; a classification of all further bases with integral powers as exactly the 3-smooth numbers; and a density dichotomy showing that the fractional parts of nonintegral solutions, if any exist, are dense in $[0, 1]$ — whence the full conjecture is *equivalent* to the existence of any nonempty open subinterval of $(0, 1)$ free of such fractional parts. We close with an assessment of the obstructions and a list of research directions.

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*Survey compiled with AI assistance (Claude Fable 5, Anthropic). Every theorem quoted below is machine-checked in Lean 4 against Mathlib in the repository <https://github.com/VladimirReshetnikov/ProveIt>; the named Lean identifiers are the authoritative statements. The local Gelfond–Schneider development is ported from Michail Karatarakis’s Mathlib formalization (Mathlib PR #42911, Apache 2.0).

1 The problem and its status

Throughout, $\theta := \log 3 / \log 2 = \log_2 3 = 1.5849625 \dots$

Conjecture 1.1 (Two-base integer-exponent conjecture; Alaoglu–Erdős). *If $x \in \mathbb{R}$ satisfies $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

Lean: `LeanProofs.TwoBaseIntegerExponent.AlaogluErdosConjecture` (recorded as a `Prop`, not asserted)

Questions of this type go back to Alaoglu and Erdős’s 1944 study of colossally abundant numbers [1], where the arithmetic nature of simultaneous rational powers p^x, q^x of distinct primes controls the quotients of consecutive colossally abundant numbers (a theme reaching back to Ramanujan’s superior highly composite numbers [2]). The conjecture is the special case $\{x_1, x_2\} = \{1, x\}$, $\{y_1, y_2\} = \{\log 2, \log 3\}$ of the *four exponentials conjecture* (Schneider’s first problem [3]; see Waldschmidt [4]), and this reduction is itself part of the corpus:

Theorem 1.2 (Reduction to four exponentials). *The real four exponentials conjecture — for any $\mathbb{Q}$ -linearly independent pairs u_1, u_2 and v_1, v_2 of reals, some $e^{u_i v_j}$ is transcendental — implies Conjecture 1.1.*

Lean: `alaogluErdosConjecture_of_realFourExponentialsConjecture`

By contrast, the analogous statement with *three* multiplicatively independent bases is within reach of the six exponentials theorem, and its special case is fully formalized by an elementary interpolation-determinant argument following Waldschmidt [5]:

Theorem 1.3 (Three-base theorem). *If $2^x, 3^x, 5^x \in \mathbb{Z}$ for a real x , then $x \in \mathbb{Z}$.*

Lean: `LeanProofs.IntegerExponent` (modules `IntegerExponent`, `DeterminantBound`, `ExponentialKernel`, `Grid`)

Everything below is an unconditional restriction on a hypothetical counterexample to Conjecture 1.1. The overall picture that emerges is a pincer: elementary and transcendence methods force any nonintegral solution to be large, transcendental, badly approximable, and confined to a rigid two-generator multiplicative lattice of asymptotic density zero, while the density dichotomy of Section 8 shows that the remaining gap — however it is measured along the fractional-part axis — is exactly the conjecture itself.

2 Notation and reformulations

For $x \geq 0$ with $2^x \in \mathbb{Z}$ write $m = 2^x \in \mathbb{N}$; then $x = \log_2 m$ and $3^x = m^\theta$. This parametrizes the problem by natural numbers:

Definition 2.1 (Candidates). A positive natural number m is a *two-base candidate* if $m^\theta \in \mathbb{Z}$.

Lean: `TwoBaseNaturalCandidate`

Powers of two are exactly the candidates corresponding to integral exponents.

Theorem 2.2 (Candidate reformulation). *Conjecture 1.1 holds if and only if every two-base candidate is a power of two.*

Lean: `alaogluErdosConjecture_iff_candidates_are_powers_of_two`

Two elementary but essential closure properties: any solution x is nonnegative; solutions are closed under addition and under multiplication by positive naturals, so the solution set is an additive submonoid of $[0, \infty)$ containing $\mathbb{N}$; and a *rational* solution is already integral, by unique factorization applied to $2^{p/q} \in \mathbb{Z}$:

Lean: `nonneg_of_two_rpow_integer, rpow_integer_add, rpow_integer_nat_mul`

Theorem 2.3. *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$ then $x \in \mathbb{Z}$. Consequently every nonintegral solution of the two integrality conditions is irrational.*

Lean: `IntegerExponent.integer_of_rational_of_two_rpow_integer, irrational_of_not_integer_of_two_rpow_integer`

3 Elementary rigidity: gaps and Diophantine lower bounds

An integral power caught strictly between consecutive integral powers of its base must clear both by at least one; normalizing gives explicit fractional-part gaps.

Theorem 3.1 (Fractional-part gaps). *Let $x \in (n, n+1)$ with $n \in \mathbb{N}$, $t = x - n$, and suppose $2^x, 3^x \in \mathbb{Z}$. Then*

$$1 + 2^{-n} \leq 2^t \leq 2 - 2^{-n}, \quad 1 + 3^{-n} \leq 3^t \leq 3 - 3^{-n}.$$

Lean: `two_three_fractional_part_gaps, two_three_rpow_gaps_near_nat`

Near the endpoints these exclusion zones have width $\asymp 2^{-n}$: they shrink as the integer part grows (the *shrinking target* phenomenon). Applied to all natural multiples kx of a solution — which are again solutions — this yields effective lower bounds on rational approximation:

Theorem 3.2 (Effective irrationality bound). *Let $x \notin \mathbb{Z}$ satisfy $2^x \in \mathbb{Z}$, and write $m = 2^x \in \mathbb{N}$. Then for all integers p and all $k \geq 1$,*

$$\left| x - \frac{p}{k} \right| \geq \frac{1}{2 k m^k}.$$

Lean: `rational_approximation_exponential_lower_bound, exists_natural_base_rational_approximation_lower_bound; sharper exact forms in ShrinkingTarget`

The bound is exponentially weak in k but completely explicit; it quantifies Theorem 2.3 and is the elementary counterpart of the (inaccessible) polynomial bounds that linear forms in logarithms would give.

4 Verified zero-free regions

Ruling out all candidates $m < M$ rules out all nonintegral solutions with $2^x < M$. For each non-power-of-two m the certificate is a pair of exact integer power comparisons against rational enclosures $L_p/L_q < \theta < U_p/U_q$ drawn from continued-fraction convergents of θ : from $a^{L_q} < m^{L_p}$ and $m^{U_p} < (a+1)^{U_q}$ one traps $a < m^\theta < a+1$, so $m^\theta \notin \mathbb{Z}$.

Theorem 4.1 (Zero-free region). *If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 4096$, then $x \in \mathbb{Z}$. Equivalently, every nonintegral solution satisfies $x \geq 12$, and every candidate that is not a power of two is at least 4096.*

Lean: `integer_of_two_three_rpow_integer_of_two_rpow_lt_4096, twelve_le_of_not_integer_of_two_three_rpow_integer, le_of_twoBaseNonintegerCandidate_4096`

Remark 4.2 (Temporary note on verification status). At the time this revision was typeset, the Lean kernel replay of the 4096-certificate table was still running; Theorem 4.1 and each downstream use of the numeral 12 should be read as pending final acceptance by Lean. The certificate table has already passed two independent exact validations (PARI/GP and Python big-integer arithmetic), and the previously accepted region $2^x < 256$ (hence $x \geq 8$) is unaffected. This remark will be removed once the build completes.

The verification history is itself of methodological interest: the bound $2^x < 10$ is proved inline; $2^x < 100$ and $2^x < 256$ use per-value `norm_num` certificates with three enclosure tiers (exponents ~ 500 , ~ 1500 , ~ 25000); the extension to 4096 verifies 3836 certificates through a single kernel-replayed `decide` over a table generated by PARI/GP and independently revalidated with exact big-integer arithmetic in Python. Five enclosure tiers suffice, the largest (convergents 176251/111202 and 301994/190537, numbers of roughly 90,000 digits) being needed for the single value $m = 2762$, whose θ -power is within $3 \cdot 10^{-9}$ of an integer in logarithmic scale. No axioms beyond the Lean kernel are involved — in particular no `native_decide`.

A descent mechanism converts the zero-free region into constraints at all heights:

Theorem 4.3 (Arithmetic descent). *If the two integral outputs factor simultaneously as $2^x = 2^r a^k$ and $3^x = 3^r b^k$ with $k \geq 1$, then $(x - r)/k$ is again a solution; combined with the zero-free region this gives the unconditional bound $r + 8k \leq x$ for nonintegral x (with the numeral 8 inherited from the $2^x < 256$ stage). In particular common perfect-power shapes of the two outputs are strongly restricted.*

Lean: `affine_descent_integral_powers`, `base_depth_add_eight_mul_degree_le`,
`min_output_base_factorization_add_eight_le`

5 Local combinatorics: progressions and zero density

The function $m \mapsto m^\theta$ is strictly convex, and its second differences at moderate steps lie strictly between 0 and 1 — hence cannot be integers.

Theorem 5.1 (AP obstruction). *Let $n, h \geq 1$ with $h^5 \leq n$. Then $n, n + h, n + 2h$ are not all two-base candidates. In particular no three consecutive naturals are candidates, and among any aligned block of three, at most two are candidates (so at most $2N$ of the first $3N$ naturals are candidates).*

Lean: `not_three_arithmetic_progression_twoBaseNaturalCandidates`,
`not_three_consecutive_logThreeDivLogTwo_rpow_integers`, `twoBaseNaturalCandidate_Icc_card_le`

The hypothesis $h^5 \leq n$ is essential to the curvature argument: for steps comparable to n the second difference is large and the obstruction disappears. (This locality is a real barrier; see Section 10.) Feeding the local obstruction into the theory of Roth numbers gives quantitative sparsity:

Theorem 5.2 (Zero density). *For every $H \geq 1$ the number of candidates below N is at most $H^5 + \lceil (N - H^5)/H \rceil r_3(H)$, where $r_3(H)$ is the maximal size of a 3-AP-free subset of $\{0, \dots, H - 1\}$. Consequently the candidate counting function is $o(N)$: the candidates (and a fortiori the values 2^x of nonintegral solutions) have asymptotic density zero.*

Lean: `twoBaseNaturalCandidatesBelow_card_le`, `twoBaseNaturalCandidatesBelow_isLittleO_id`,
`tendsto_twoBaseNonintegerCandidatesBelow_density_zero`

6 Multiplicative structure: rank two and the odd core

Transcendence input (Section 7) upgrades density zero to logarithmic counting. The mechanism is linear algebra on prime-exponent vectors: the vectors of all candidates span a rational plane containing the vector of 2.

Theorem 6.1 (Rank two). *The $\mathbb{Q}$ -span of $\{v(m) : m \text{ candidate}\} \subset \bigoplus_p \mathbb{Q}$, where $v(m)$ is the prime-factorization vector, has dimension at most 2. Consequently the number of candidates below N is at most $(\log_2 N)^2$.*

Lean: `finrank_span_primeExponentVectors_le_two, twoBaseNaturalCandidatesBelow_card_le_clog_sq`

Pairwise, any two candidates satisfy exact power relations modulo powers of two, and the odd-prime exponent profiles of all candidates are mutually proportional. These pairwise relations glue globally:

Lean: `fixed_power_relation_of_odd_prime_factor, odd_factorization_cross_mul`

Theorem 6.2 (Common odd core). *There is a single odd $w > 1$ such that every two-base candidate equals $2^i w^j$ for some $i, j \in \mathbb{N}$; candidates that are not powers of two have $j \geq 1$. In exponent coordinates: every solution of the two integrality conditions has the form*

$$x = i + j \log_2 w, \quad i, j \in \mathbb{N}.$$

(If the conjecture holds, any odd w serves vacuously with $j = 0$; otherwise w is the primitive base of the odd part of any candidate that is not a power of two.)

Lean: `exists_common_odd_core, exists_common_odd_core_split, exists_common_odd_core_exponent`

Corollary 6.3 (Improved counting). *The number of candidates below N is at most $\log_2 N \cdot \log_3 N$, sharpening Theorem 6.1 by the factor $\log_2 3$. Moreover each dyadic block $[2^T, 2^{T+1})$ contains at most $\log_3(2^{T+1}) \approx 0.631(T+1)$ candidates: the number of solutions with $x \in [T, T+1)$ grows at most linearly in T , against the quadratic global bound.*

Lean: `twoBaseNaturalCandidatesBelow_card_le_clog_mul_clog, twoBaseNonintegerCandidatesBelow_card_le_clog_mul_clog, twoBaseNaturalCandidatesInDyadicBlock_card_le`

The logarithmic-square order cannot be improved by counting alone: if a single candidate with an odd prime factor exists, multiplicative closure generates injective boxes $2^i a^j$, so the count below $2^k a^l$ is at least kl — a conditional lower bound of exactly the same quadratic-in-logarithm shape.

Lean: `twoBaseNaturalCandidatesBelow_card_ge_box, twoBaseNonintegerCandidatesBelow_card_ge_box`

7 Transcendence

The corpus contains a complete Lean proof of the Gelfond–Schneider theorem (Hilbert’s seventh problem), following Hua’s exposition [6] and ported from Michail Karataakis’s Mathlib formalization.

Theorem 7.1 (Gelfond–Schneider). *If α, β are algebraic, $\alpha \neq 0, 1$, and β is irrational, then α^β is transcendental.*

Lean: `GelfondSchneider.transcendental_cpow_of_isAlgebraic_of_irrational`

Corollary 7.2 (Algebraic–integral dichotomy). *If $2^x \in \mathbb{Z}$ then x is either an integer or transcendental; being algebraic is equivalent to being an integer for such x . In particular every nonintegral solution of the two-base conditions is transcendental, and $\theta = \log_2 3$ itself is transcendental.*

Lean: `transcendental_of_not_integer_of_two_rpow_integer, isAlgebraic_iff_integer_of_two_rpow_integer, transcendental_of_not_integer_of_two_three_rpow_integer, transcendental_logThreeDivLogTwo`

A further rigidity statement classifies all bases that could join a counterexample:

Theorem 7.3 (3-smooth classification). *Suppose $x \notin \mathbb{Z}$, $2^x \in \mathbb{Z}$, $3^x \in \mathbb{Z}$. Then for a positive natural a , the power a^x is an integer if and only if $a = 2^u 3^v$ is 3-smooth. In particular the presence of any third base with a prime factor other than 2 and 3 forces $x \in \mathbb{Z}$.*

Lean: `rpow_integer_iff_eq_two_pow_mul_three_pow_of_not_integer,`
`integer_of_two_three_a_rpow_integer_of_prime_factor`

Theorem 7.3 is best possible in the sense that its “only if” direction for 3-smooth a is genuinely available to a counterexample: $(2^u 3^v)^x$ is then an integer automatically.

8 The fractional-part dichotomy

Since natural multiples of a solution are solutions, one nonintegral solution generates infinitely many, with irrational generator; Dirichlet approximation and a rotation-stepping argument then spread their fractional parts densely.

Theorem 8.1 (Density dichotomy). *Suppose some $x \notin \mathbb{Z}$ satisfies $2^x, 3^x \in \mathbb{Z}$. Then for every $t \in [0, 1]$ and every $\varepsilon > 0$ there is a nonintegral solution y with $|\text{frac}(y) - t| < \varepsilon$. Moreover the set of nonintegral solutions is infinite.*

Lean: `dense_fract_of_noninteger_solution,` `infinite_noninteger_solutions_of_exists,`
`exists_pos_nat_fract_close_of_irrational`

Theorem 8.2 (Gap equivalences). *Each of the following is equivalent to Conjecture 1.1:*

1. *some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part of a nonintegral solution;*
2. *the fractional parts of nonintegral solutions are bounded away from 1;*
3. *the fractional parts of nonintegral solutions are bounded away from 0.*

Lean: `alaogluErdosConjecture_iff_fract_avoids_interval,`
`alaogluErdosConjecture_iff_fract_bounded_away_one,`
`alaogluErdosConjecture_iff_fract_bounded_away_zero`

Remark 8.3 (Consistency pincer). Theorem 3.1 bounds $\text{frac}(x)$ away from 0 and 1 by margins $\asymp 2^{-\lfloor x \rfloor}$ that decay with the height of the solution, while Theorem 8.1 produces fractional parts arbitrarily close to 0 and 1 along multiples of unbounded height. The two results meet edge-to-edge without contradiction, and Theorem 8.2 shows the decay in Theorem 3.1 is essential: *any* height-independent gap, however small and wherever located, is exactly as strong as the full conjecture. In this precise sense the elementary gap bounds are of optimal shape, and the fractional-part axis measures the entire remaining distance to Conjecture 1.1.

9 Summary table

Restriction on a nonintegral solution x	Statement	Module
irrational, in fact transcendental	Thm. 2.3, Cor. 7.2	Transcendence
$x \geq 12$, i.e. $2^x \geq 4096$	Thm. 4.1	FiniteCheck4096
$2^x = 2^i w^j$, one common odd w	Thm. 6.2	OddCore
$x = i + j \log_2 w$, $i, j \in \mathbb{N}$	Thm. 6.2	OddCore
$\leq \text{clog}_2 N \cdot \text{clog}_3 N$ values $2^x \leq N$	Cor. 6.3	OddCore
$\lesssim 0.631 T$ solutions in $[T, T+1)$	Cor. 6.3	OddCore
density zero among naturals	Thm. 5.2	ZeroDensity
no 3-AP of candidates, $h^5 \leq n$	Thm. 5.1	ZeroDensity
$ x - p/k \geq (2k m^k)^{-1}$	Thm. 3.2	ShrinkingTarget
fractional-part gaps $\asymp 2^{-\lfloor x \rfloor}$	Thm. 3.1	TwoBaseIntegerExponent
extra integral bases are 3-smooth	Thm. 7.3	ThreeSmooth
descent $r + 8k \leq x$	Thm. 4.3	ArithmeticRigidity
fractional parts dense (dichotomy)	Thm. 8.1	FractionalPartDensity
gap $\Leftrightarrow$ conjecture	Thm. 8.2	FractionalPartDensity

10 Obstructions and future directions

Where the wall is

The smallest structural question already meets an open transcendence problem. A 3-smooth number $m = 2^a 3^b$ ($b \geq 1$) is a candidate if and only if $3^a E^b \in \mathbb{Z}$, where

$$E = 3^{\log_2 3} = e^{(\log 3)^2 / \log 2} = 5.7045\dots,$$

and even the irrationality of E is open: it is a “three-logarithms” quantity outside the reach of Gelfond–Schneider, Baker’s method, and the six exponentials theorem. Thus no currently known technique can exclude odd core $w = 3$, which is why the corpus emphasizes counting, structure, and verified finite regions rather than infinite families of exclusions.

Similarly, the AP obstruction (Theorem 5.1) is intrinsically *local*: the curvature argument fails for steps h comparable to n . A *global* AP obstruction would be genuinely new and strong — for instance, it would forbid candidates with odd part of the form $2^s \pm 1$, via progressions such as $(2^i, 2^{i+t-1}, 2^i(2^t - 1))$. Any method controlling three-term progressions of candidates at commensurate scales is therefore a priority target.

Concrete directions

1. **Equidistribution.** Upgrade the density dichotomy to Weyl equidistribution of the multiples kx_0 modulo one (not yet in Mathlib), and further to discrepancy bounds (Erdős–Turán). This would make Theorem 8.2 quantitative: an interval of width δ free of fractional parts could be traded against solution counts up to an explicit height $H(\delta)$, tying Sections 4 and 8 together.
2. **Five exponentials.** The five exponentials theorem (Waldschmidt) applied to $\{1, x\}$, $\{\log 2, \log 3\}$ yields: for any nonintegral solution x and any nonzero algebraic γ , the number $e^{\gamma x}$ is transcendental. Formalizing it (or the sharp six exponentials theorem of Roy) would add a new axis of restrictions; the effort is comparable to the completed Gelfond–Schneider port.
3. **Effective irrationality measures for θ .** Padé/hypergeometric methods (Rhin-style) give effective irrationality measures for $\log_2 3$. Formalized, they would replace the per-value certificate tiers of Section 4 by a uniform polynomial criterion, likely extending

the zero-free region by orders of magnitude, and would give polynomial (rather than exponential) lower bounds in Theorem 3.2 for the specific generator $\log_2 w$.

4. **Larger verified regions.** The kernel-replayed certificate table scales: the range $[4096, 16384)$ needs roughly twenty values whose enclosures require convergent denominators $\sim 10^7$ (integers of $\sim 10^7$ digits) — feasible on a machine with more memory, mechanical otherwise. Improving the descent constant of Theorem 4.3 from 8 to 12 using the new region is immediate bookkeeping.
5. **Sharper local combinatorics.** The AP hypothesis $h^5 \leq n$ can be relaxed toward $h \leq n^{(2-\theta)/2-\varepsilon}$ ($(2-\theta)/2 \approx 0.2075$), and longer-progression or higher-difference obstructions (Δ^k of m^θ for $k \geq 3$ lies in $(0, 1)$ on wider ranges) would improve the Roth-number density rate and the block counting of Corollary 6.3.
6. **Structure of the exponent semigroup.** By Theorem 6.2 the nonintegral solutions form $\{i + j \log_2 w : (i, j) \in B\}$ where the set of occurring j 's is closed under addition. Numerical semigroup structure (Frobenius-type gaps) combined with the dyadic block bound might force either $B = \emptyset$ or contradictory growth — a purely combinatorial attack surface that has not been exploited.
7. **Colossally abundant interface.** Formalize the Alaoglu–Erdős application: Conjecture 1.1 (in its rational-power form) implies that quotients of consecutive colossally abundant numbers are prime. This would connect the corpus to its historical source and to Ramanujan's superior highly composite numbers.

References

- [1] L. Alaoglu and P. Erdős, *On highly composite and similar numbers*, Trans. Amer. Math. Soc. **56** (1944), 448–469.
- [2] S. Ramanujan, *Highly composite numbers*, Proc. London Math. Soc. (2) **14** (1915), 347–409; and the annotated “Highly composite numbers” manuscript, Ramanujan J. **1** (1997), 119–153.
- [3] Th. Schneider, *Einführung in die transzendenten Zahlen*, Springer, 1957.
- [4] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups*, Grundlehren der mathematischen Wissenschaften 326, Springer, 2000.
- [5] M. Waldschmidt, *Six exponentials theorem — irrationality*, expository note.
- [6] L.-K. Hua, *Introduction to Number Theory*, Springer, 1982, Chapter 17.9.

A Formalization inventory

All modules live under `Analysis/ExponentialIdentities/Lean/ExponentialIdentities/`, in namespace `LeanProofs`. Lean toolchain 4.31.0, Mathlib v4.31.0. No `sorry`, no extra axioms, no `native_decide`; the certificate tables are replayed by the kernel.

Module	Contents
<code>IntegerExponent(...)</code>	three-base theorem via interpolation determinants
<code>TwoBaseIntegerExponent</code>	conjecture statement, candidates, gaps, $2^x < 10$
<code>.../FiniteCheck, FiniteCheck256</code>	zero-free regions $2^x < 100$, $2^x < 256$
<code>.../FiniteCheck4096</code>	zero-free region $2^x < 4096$ (kernel <code>decide</code> table)
<code>.../ZeroDensity</code>	AP obstruction, Roth-number counting, $o(N)$
<code>.../NonintegerDensity</code>	nonintegral candidate refinement, density zero
<code>.../MultiplicativeRank</code>	$\text{rank} \leq 2$, θ irrational, $(\text{clog}_2 N)^2$
<code>.../CandidateStructure</code>	pairwise power relations, conditional box growth
<code>.../OddCore</code>	common odd core, $\text{clog}_2 N \text{ clog}_3 N$, dyadic blocks
<code>.../ArithmeticRigidity</code>	descent, $r + 8k \leq x$
<code>.../ThreeSmooth</code>	3-smooth classification of further bases
<code>.../ShrinkingTarget</code>	fract shrinking targets, Diophantine lower bounds
<code>.../Transcendence</code>	GS applications: transcendence of solutions, of $\log_2 3$
<code>.../FractionalPartDensity</code>	density dichotomy, gap equivalences, infinitude
<code>.../FourExponentialsReduction</code>	$4\text{EC} \Rightarrow$ conjecture
<code>GelfondSchneider/...</code>	Gelfond–Schneider theorem (port of Karataarakis)